Problem Set 4

Problem 1. Find the unit tangent $\mathbf{T}(t)$, unit normal $\mathbf{N}(t)$, Binormal $\mathbf{B}(t)$ of the following curves at a given point.

(i) $\mathbf{r}(t) = (\cos t,\ t,\ \sin t)$, $t = 0$.

(ii) $\mathbf{r}(t) = (\cos t,\ \sin t,\ \ln \cos t)$, $t = 0$.

(iii) $\mathbf{r}(t) = (t,\ \frac{1}{2}t^2,\ t^2)$, $t = 0$.

Problem 2. (Limit).

(i) Consider the following function $f : \mathbb{R}^2 \to \mathbb{R}$ given as

$$f(x, y) = \begin{cases} 2 & \text{if } y = 2x \\ 0 & \text{otherwise.} \end{cases}$$

For $(0, 0)$, $(1, 1)$ explore the following:

(a) whether limit exists along the line parallel to x-axis.

(b) whether limit exists along the line parallel to y-axis.

(c) whether limit exists along the line line parallel to $y = mx$ [for different values of m (need to be chosen so that given point lies on corresponding line)]

(ii) Construct examples for the following:

(a) $f : \mathbb{R}^3 \to \mathbb{R}$ such that at origin limit exists along x-axis but does not exist along y-axis and vice versa.

(b) $f : \mathbb{R}^3 \to \mathbb{R}$ such that at origin limit exists along x-axis as well as y-axis but does not exist along $y = mx$ (for some particular value of m).

(c) $\star$ $f : \mathbb{R}^3 \to \mathbb{R}$ such that at origin limit exists along x-axis as well as y-axis and along $y = mx$ (for all values of m). But limit does not exist (or gives a different value) along some curve.

(iii) Consider the following function $f : \mathbb{R}^2 \to \mathbb{R}$ given as

$$f(x, y) = \begin{cases} 2 & \text{if } y = x^2 \\ 0 & \text{otherwise.} \end{cases}$$

For different points on domain explore the following

(a) whether limit exists along the line parallel to x-axis.

(b) whether limit exists along the line parallel to y-axis.

(c) whether limit exists along the line line parallel to $y = mx$ [for different values of m (need to be chosen so that given point lies on corresponding line)]

(iv) Find the limit, if it exists, or show that the limit does not exist.

(a) $\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}}$

(b) $\lim_{(x,y)\to(0,0)} \frac{x^4+y^4}{x^2+y^2}$

(c) $\lim_{(x,y)\to(0,0)} \frac{x^2ye^y}{x^4+4y^2}$

(v) Consider

$$f(x,y) = \frac{x^2-y^2}{x^2+y^2} \text{ if } (x,y) \neq 0.$$

(a) Examine whether limit exists when you approach $(0,0)$ along the line $y = mx$ (i.e., you need to compute $\lim_{h\to 0} f(h, mh)$).

(b) Can you define the value of f at $(0,0)$ to make it continuous.

(vi) $\star$ Can you construct an example such that both the below limit exist and

$$\lim_{x\to 0}\lim_{y\to 0} f(x,\, y) \neq \lim_{y\to 0}\lim_{x\to 0} f(x,y).$$

Is it true that if limit exists at $(0,\, 0)$ then we must have

$$\lim_{x\to 0}\lim_{y\to 0} f(x,\, y) = \lim_{y\to 0}\lim_{x\to 0} f(x,y).$$

Problem 3. (Continuity)

(i) Show that the function $f : \mathbb{R}^m \to \mathbb{R}$ such that $f(\mathbf{x}) = ||\mathbf{x}||$ is continuous.

(ii) Show that the functions $f_i : \mathbb{R}^m \to \mathbb{R}$ such that $f_i(\mathbf{x}) = \mathbf{x}(i)$ are continuous for each $1 \leq i \leq m$.

(iii) Let $\mathbf{c} \in \mathbb{R}^m$, Show that the function $f : \mathbb{R}^m \to \mathbb{R}$ given by $f(\mathbf{x}) = \mathbf{c} \cdot \mathbf{x}$ is continuous.

(iv) Consider $f : \mathbb{R} \to \mathbb{R}^2$ such that $f(x) = (f_1(x), f_2(x))$ such that both $f_1 : \mathbb{R} \to \mathbb{R}$ and $f_2 : \mathbb{R} \to \mathbb{R}$ are continuous. Then show that f must be continuous too.

(*v*) Consider continuous maps $f_1, f_2 : \mathbb{R} \to \mathbb{R}$ and define $f : \mathbb{R}^3 \to \mathbb{R}^2$ such that $f(x, y, z) = (f_1(z), f_2(x))$.

(*vi*) For any subset $A \subset \mathbb{R}^m$. Consider continuous functions $f, g : A \to \mathbb{R}$. Suppose $\mathbf{x}_0 \in A$ such that $g(\mathbf{x}_0) \neq 0$ then show that there exists $\varepsilon \geq 0$ such that $g(x) \neq 0$ for all $\mathbf{x} \in B_\varepsilon(\mathbf{x}_0)$. Under the above conditions show that $f/g : A \to \mathbb{R}$ is continuous at x_0.

(*vii*) Determine the set of points at which the function is continuous.

(a) $f : \mathbb{R}^3 \to \mathbb{R}^3$ defined as $f(x, y, z) = (\sin(x^2 + 5yz),\ \sqrt{1 - \cos^2 x},\ e^{yz})$.

(b) $f : \mathbb{R}^3 \to \mathbb{R}$ defined as $f(x, y, z) = \frac{x^2+y^2+\log(x^2+y^2)}{x+y+z-1}$

(c) $f : \mathbb{R}^2 \to \mathbb{R}$ defined as

$$f(x, y) = \begin{cases} \frac{xy}{x^2+xy+y^2} & \text{if } (x,\ y) \neq 0 \\ 0 & \text{if } (x,\ y) = 0 \end{cases}$$

Problem 4. (Partial derivatives)

(*i*) Find all 1st order partial derivatives for the following function:

(a) $f(x,\ y) = \int_x^y \cos(e^t)dt$

(b) $f(x,\ y) = e^{xy}\sin y$

(c) $f(x,\ y,\ z) = \sqrt{\sin^2 x + \sin^2 y + \sin^2 z}$

(d) $f(x,\ y,\ z) = x^a y^b z^c$

(*ii*) Find the 2nd order partial derivatives f_{xy} and f_{yx} of function f whenever they exist and check whether they are equal.

(a) $f(x,\ y) = e^{xy}\sin y$

(b) $f(x,\ y) = x^2 + y^2\sin(xy)$

(c) $f(\mathbf{x}) = \sum_{i=1}^n \sum_{j=1}^n a_{ij}x_ix_j$ with $a_{ij} = a_{ji}$, $1 \leq i,\ j \leq n$

(*iii*) Construct examples for the following:

(a) $f : \mathbb{R}^2 \to \mathbb{R}$ such that at origin f_{xy} and f_{yx} exists but not equal.

(b) $f : \mathbb{R}^2 \to \mathbb{R}$ such that at origin f_{xy} and f_{yx} exists, equal and continuous.

(c) $\star$ $f : \mathbb{R}^2 \to \mathbb{R}$ such that at origin f_{xy} and f_{yx} exists and equal but not continuous.

Problem 5. (Directional derivative)

(*i*) Consider the function $f : \mathbb{R}^3 \to \mathbb{R}$ defined as $f(\mathbf{x}) = ||\mathbf{x}||^2$ and check the following:

(a) Check whether all partial derivative exists at origin or not and if yes then find it's value.

(b) Check whether all partial derivative exists at any non zero vector $\mathbf{x}$ or not and if yes then find it's value.

(c) Fix unit vector $\mathbf{u}$ and Check whether the directional derivative exists at origin or not and if yes then find it's value.

(*ii*) Fix $\mathbf{a} \in \mathbb{R}^3$ and consider the function $f : \mathbb{R}^3 \to \mathbb{R}$ defined as $f(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x}$ and check the following:

(a) Check whether all partial derivative exists at origin or not and if yes then find it's value.

(b) Check whether all partial derivative exists at any non zero vector $\mathbf{x}$ or not and if yes then find it's value.

(c) Fix unit vector $\mathbf{u}$ and Check whether the directional derivative exists at origin or not and if yes then find it's value.

(*iii*) Consider the function

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{if } (x,\, y) \neq (0, 0) \\ 0 & \text{if } (x,\, y) = (0, 0) \end{cases}$$

Then check the following:

(a) Check whether all partial derivative exists at origin or not and if yes then find it's value.

(b) Check whether all partial derivative exists at any non zero vector $\mathbf{x}$ or not and if yes then find it's value.

(c) for any unit vector $\mathbf{u}$ and check whether the directional derivative exists at origin or not and if yes then find it's value.

(d) Show that f is not continuous at origin.

(*iv*) Consider the function

$$f(x, y) = \begin{cases} \frac{xy^2}{x^2+y^4} & \text{if } (x,\, y) \neq (0, 0) \\ 0 & \text{if } (x,\, y) = (0, 0) \end{cases}$$

Then check the following:

(a) Check whether all partial derivative exists at origin or not and if yes then find it's value.

(b) for any unit vector $\mathbf{u}$ and check whether the directional derivative exists at origin or not and if yes then find it's value.

(c) Show that f is not continuous at origin.